

Regulated Credit Scoring with AMEX Default Prediction

1. Executive Summary

This project develops a credit default prediction model using the American Express Default Prediction dataset. The analysis compares a last-statement baseline model, a customer-level aggregated feature model, and a monotonic LightGBM model designed with domain-inspired constraints.

The aggregated LightGBM model achieved a ROC-AUC of 0.9610, a PR-AUC of 0.8974, and an F1-score of 0.8137. SHAP analysis was used to explain how individual features affected predicted default risk.

2. Business Problem

Credit scoring models estimate the probability that a customer will default. In regulated financial environments, predictive performance alone is not sufficient. Models should also produce stable, explainable, and economically reasonable predictions.

The goal of this project is to evaluate whether customer-level historical features improve predictive performance and whether monotonic constraints can improve alignment with credit risk intuition.

3. Dataset and Preprocessing

The dataset contains approximately 5.53 million monthly statement records for 458,913 customers.

Each customer has multiple historical statement rows. The target variable is binary:

- target = 1: default
- target = 0: non-default

The following preprocessing steps were applied:

- Converted S_2 into a datetime variable
- Sorted records by customer and statement date

- Encoded categorical features
- Split the customer-level dataset into 80% training and 20% validation sets
- Preserved the class distribution using stratified sampling

4. Modeling Approach

4.1 Last Statement Baseline

The baseline model uses only the most recent statement for each customer. Logistic Regression and LightGBM were trained to establish a performance benchmark.

4.2 Aggregated Features Model

The aggregated model summarizes each customer's complete statement history. For numerical variables, the following statistics were calculated:

- Mean
- Standard deviation
- Minimum
- Maximum
- Last value

Categorical variables were represented using their latest values. A statement count feature was also included. This process created 897 customer-level features.

4.3 Monotonic LightGBM

The monotonic model applies domain-inspired directional constraints:

- D_* variables: positive relationship with default risk
- P_* variables: negative relationship with default risk
- Standard deviation and categorical variables: unconstrained
- Other variables: unconstrained

Because the AMEX variables are anonymized, these directions should be interpreted as approximate assumptions rather than confirmed causal relationships.

5. Model Results

Model	ROC-AUC	PR-AUC	Precision	Recall	F1
Last Statement LightGBM	0.9601	0.8952	0.7231	0.9150	0.8078
Aggregated LightGBM	0.9610	0.8974	0.7390	0.9053	0.8137
Monotonic LightGBM	0.9605	0.8960	0.7378	0.9037	0.8124

The aggregated feature model achieved the strongest overall performance. Compared with the last-statement baseline, it improved ROC-AUC, PR-AUC, and F1-score.

The monotonic model produced slightly lower predictive performance than the unconstrained model. However, the relatively small performance decrease suggests that directional constraints can be introduced without substantially reducing model quality.

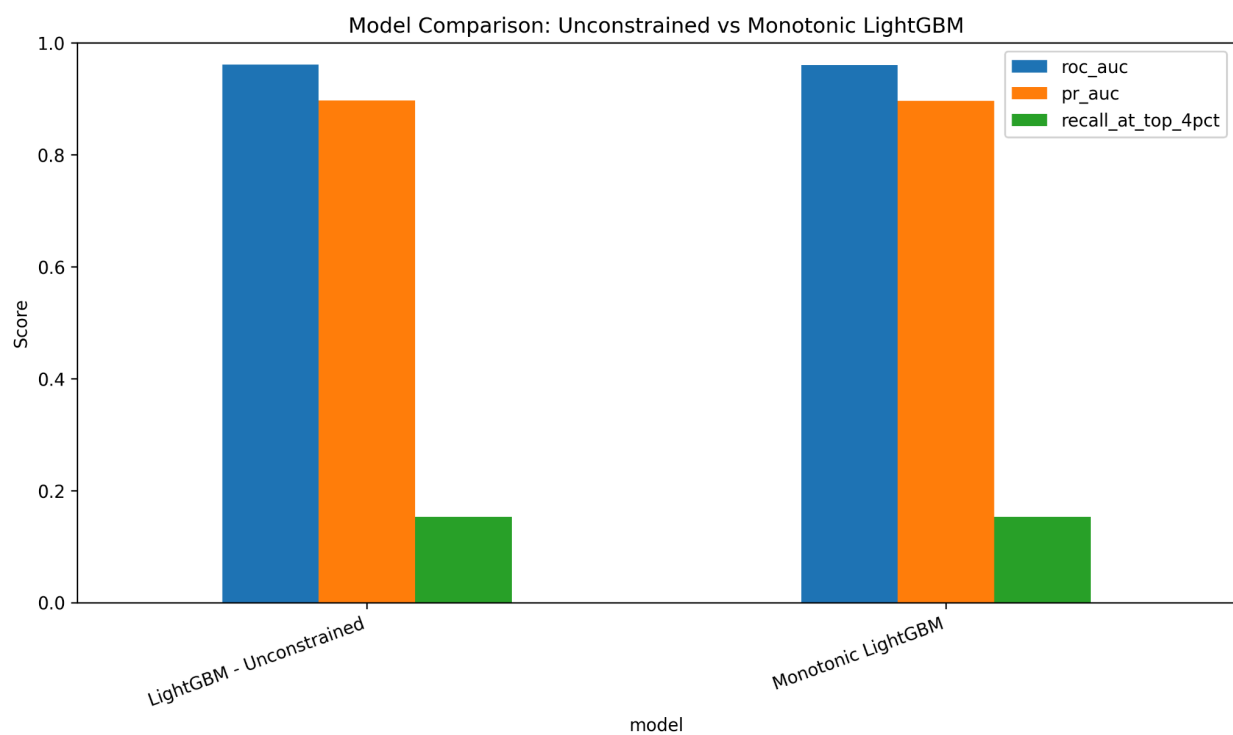


Figure 1. Performance comparison between unconstrained and monotonic LightGBM

6. Feature Importance

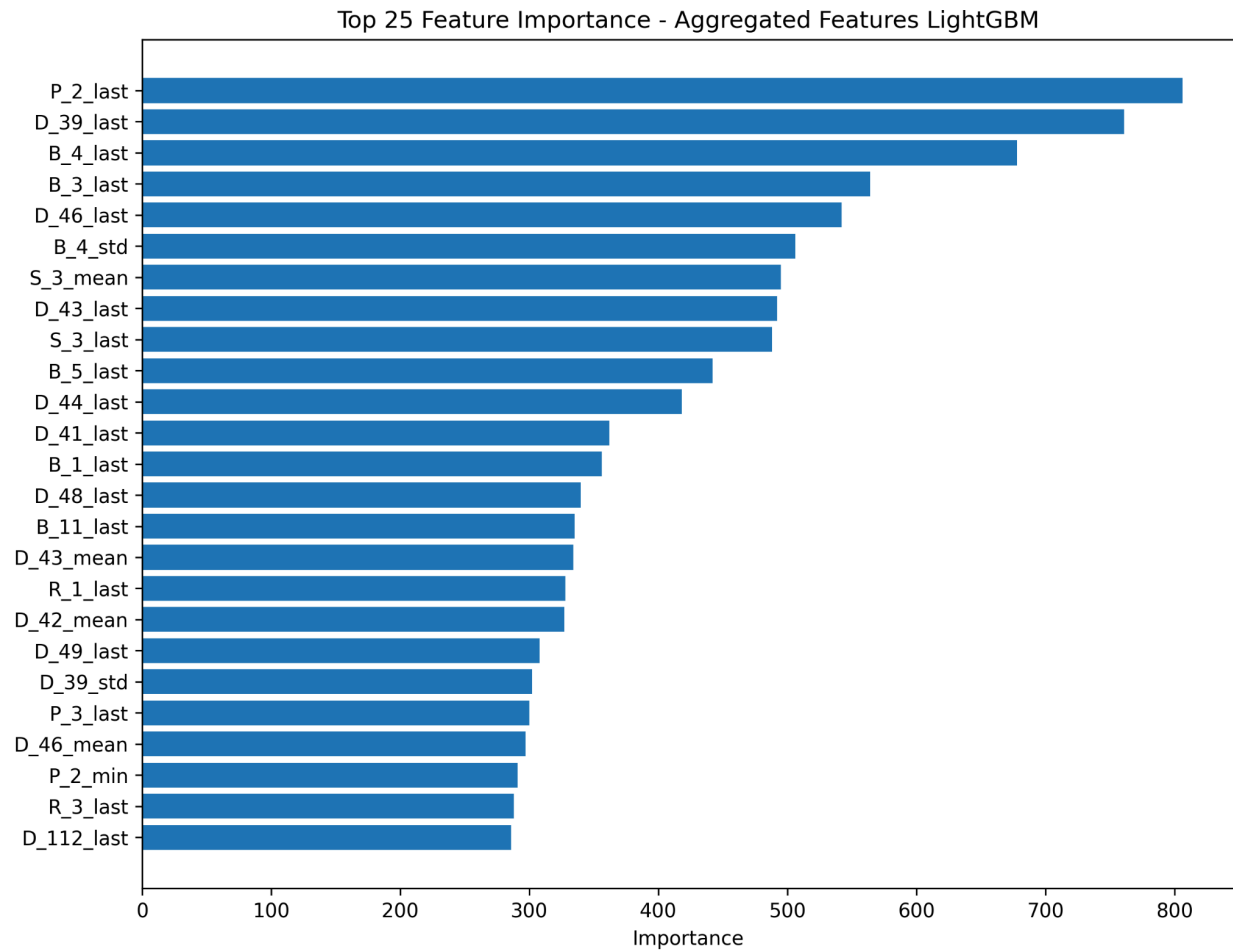
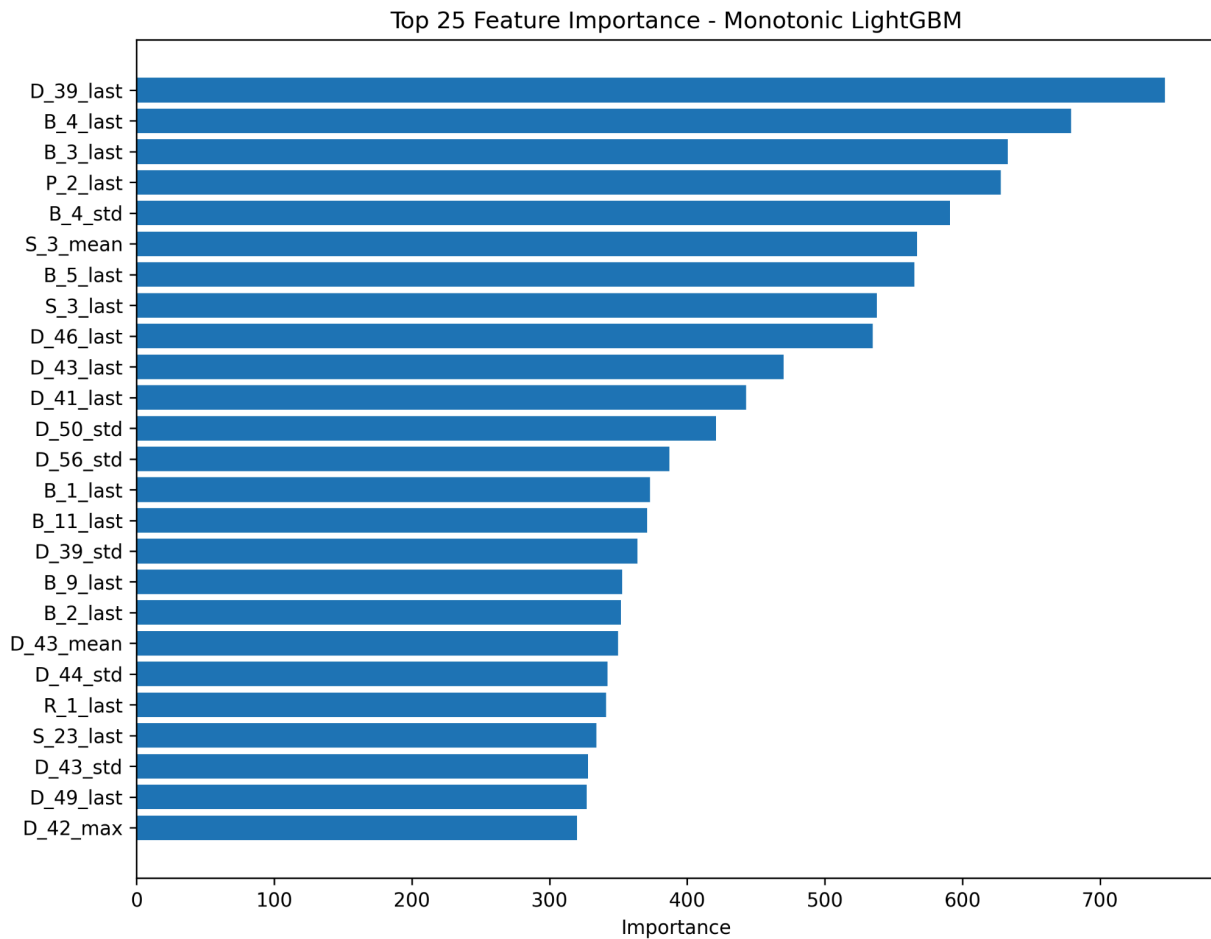


Figure 2. Top 25 features from the aggregated LightGBM model

The most important features included P_2_last, D_39_last, B_4_last, B_3_last, and D_46_last. Features based on the most recent statement appeared frequently among the highest-ranked variables, while historical summary statistics such as B_4_std and S_3_mean also contributed to the model.

7. Monotonic Constraints



The monotonic model retained many of the important variables identified by the unconstrained model. D_39_last received a positive constraint, while P_2_last received a negative constraint.

The constrained model achieved a ROC-AUC of 0.9605, compared with 0.9610 for the unconstrained model. This represents only a small reduction in discriminative performance.

8. SHAP Explainability

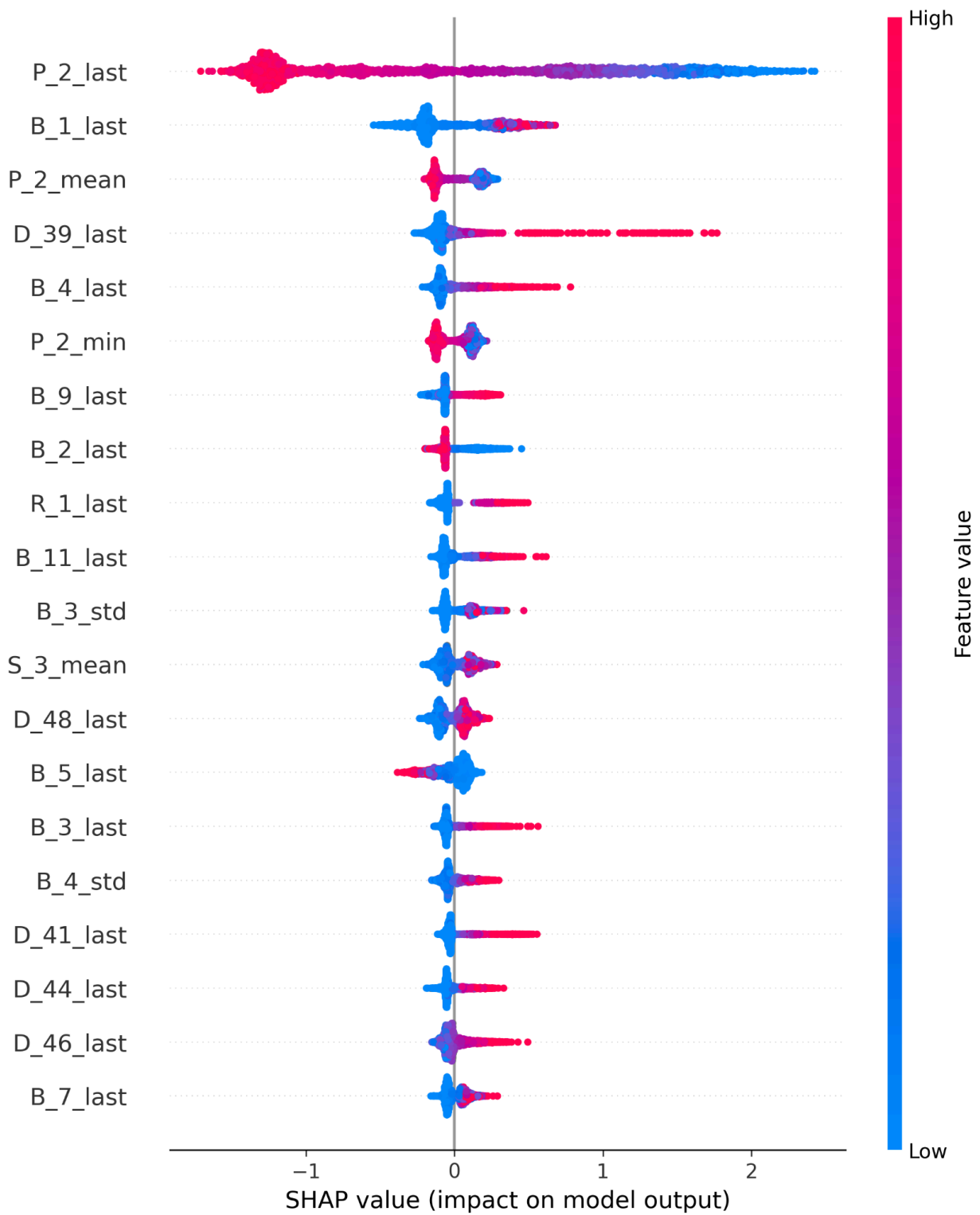


Figure 3. SHAP summary plot for the aggregated LightGBM model

The SHAP summary plot shows both feature importance and the direction of each feature's effect.

Higher values of P_2_last generally produced negative SHAP values, reducing predicted default risk. Lower P_2_last values tended to increase predicted risk.

Higher values of variables such as D_39_last and B_4_last generally increased the model's predicted default risk.



Figure 4. Example of an individual customer-level SHAP explanation.

The force plot provides a local explanation for one customer. Features shown in one direction increased predicted default risk, while features in the opposite direction reduced the prediction.

9. Limitations

This project has several limitations.

First, the AMEX variables are anonymized, so their exact financial meanings are unavailable. The monotonic constraint directions are therefore based on feature prefixes and credit-risk intuition rather than confirmed business definitions.

Second, the models were evaluated using a single train-validation split. Future work could use cross-validation or an out-of-time validation set.

Third, the analysis focuses on predictive performance and explainability. A production credit model would also require fairness testing, stability monitoring, calibration analysis, and regulatory validation.

10. Conclusion

Customer-level historical aggregation improved performance compared with using only the latest statement. The aggregated LightGBM model achieved a ROC-AUC of 0.9610, a PR-AUC of 0.8974, and an F1-score of 0.8137.

The monotonic model produced similar performance while enforcing directional relationships inspired by credit-risk principles. SHAP analysis provided both global and customer-level explanations of model predictions.

The project demonstrates how predictive performance, domain constraints, and explainability can be combined in a regulated credit-scoring workflow.